

**Teltech Acquires Confide**

IAC's Teltech has acquired the encrypted messaging app, notorious for being used in the White House. <http://digit.in/feb21-49>

**Phone Numbers On Sale**

A database of over 500 million Facebook users' phone numbers is up for sale on the Telegram app. <http://digit.in/feb21-50>

Emote Electric's
geared bike

NEXUS STARTUPS

Pushing the next big idea to market

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exus is a startup hub, that is a collaboration between the US Embassy, New Delhi and the Alliance For

Commercialization and Innovation Research. With collaborations with local organisations, access to networks of investors, training from commercialisation experts, and an extensive mentor network, the initiative has grown to become one of the top incubation programs in the country. The priority is to pick an interesting concept and help it reach the market as fast as possible. And these concepts are interesting, often not attempted elsewhere. Think geared electric bikes, surveillance systems without cameras, and solar

panels that track the Sun without moving. Here we showcase interactions with some of the most interesting startups associated with Nexus.

EMOTE ELECTRIC MAKES GEARED ELECTRIC BIKES

Founded in 2011 as eMotion Motors, Emote Electric aims to make electric bikes desirable by making them as powerful and fast as possible. Most interestingly, they are among the first in the world to produce geared electric bikes, to untap the possibilities offered. Here is an interaction with a spokesperson.

The first thing we wanted to know was the benefits offered by a geared electric bike. Emote Electric explains, "contrary to expectations, we get better efficiency by adding a gearbox to the drivetrain. The additional mechanical losses due to the gearbox are in the region of 3-4%, however the motor efficiency gains achieved by running the motor in its most efficient rpm range for 4x longer (i.e. in the correct gear for the speed), can be

as high as 35%. Other similar scooters and motorcycles that are available or being developed contain "No" gears, though in reality they have 1 reduction gear set. With only one gear, we would have a peak efficiency point at a certain speed; and have to accept poor efficiency at slow speeds, resulting in poor range in urban conditions." The approach allows Emote Electric to use a more compact motor, "by having a gearbox, we do not have to accept a trade-off between acceleration and top speed. Other companies overcome this issue by upsizing the motor. A more powerful motor requires more powerful batteries and controller (drive unit). These three put together make up two-thirds of the bill of materials of an EV. By not having this trade-off, we can use a smaller motor, controller (drive unit) and battery - while also reducing the wear and tear on the battery resulting in longer life. Our closest competitor in performance terms uses a motor that is 2.5 times larger". There is another reason that might sound